

Sunrise Report for Conway, AR During 2024

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Abstract

This paper analyzes the collection of times of each sunrise during the year 2024 in the town of Conway Arkansas. With this data we can compare the average sunrise of each month during that year. Doing this gives us a better understanding of the world we live in.

1 Introduction

Each year data is collected on sunrise times for a variety of reasons such as weather forecasting, calculating Earth Rotation speed, and for agriculture welfare. This simple project used the data for the sunrise of each day in Conway, Arkansas during the year 2024.

2 Analysis

2.1 Data Collection

Data found through (timeanddate.com, 2024) had to be processed before finding the averages of each month could be found. Due to the data being represented as a series of string in columns in a text file, all values had to be converted into integers. This was done by using python code to take the file, and separate each the string by the colon use in the values collected for times.

2.2 Data Analysis

With sunrise times from all 365 days in 2024 available the averages of each month were taken. This was done using the general formula for calculate mean.
$$\text{Average} = \frac{\text{Sum of times of each sunrise in the month}}{\text{Number of days in the month}}$$

Once this was done for each month the twelve calculated averages were plotted on a graph with months over 1.

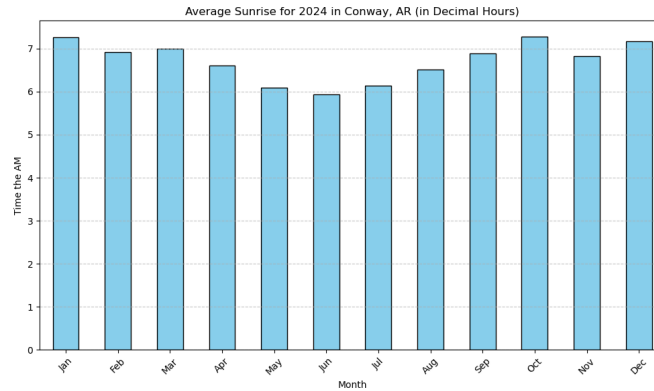


Figure 1: Graph of the sunrise averages for each month in the year 2024

3 Conclusion

3.1 Results

From the graph 1 there is a noticeable dip that appears in the middle of the year. This is due to the rolling back an hour that is done around the United States for daylight savings.

References

timeanddate.com. (2024). *Conway, arkansas, usa-sunrise, sunset, and daylength, 2024*. Retrieved April 29, 2025, from <https://www.timeanddate.com/sun/@4106458?month=4&year=2024>